PROJECT SYNOPSYS

OXYWALL - A LIVING GREEN WALL FOR CLEAN AIR

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PROBLEM:

In many schools, classrooms are often closed spaces with very little fresh air flowing in. This stale environment can make students feel drowsy, cause headaches, and even lead to frequent sickness. When dozens of people share the same air, carbon dioxide levels rise — and that's bad news for our health. Fresh air isn't just pleasant, it's essential for keeping minds alert and bodies healthy**.**

PROJECT IDEA:

OxyWall — a special algae-based paint.

This paint is eco-friendly and contains living algae that absorb carbon dioxide and release oxygen, just like plants do!

When we paint OxyWall on classroom walls, it purifies the air, making it fresher and healthier. It works like a green wall that breathes.

MATERIALS:

- Dried algae powder
- Natural binder (like cornstarch paste)
- Water
- Natural preservative (optional – like clove oil or vinegar)





Steps:

1. Preparing Algae powder:

Collect some fresh Algae, dry it and grind dry algae in fine powder.

2. Make Natural Binder:

Mix cornstarch with water. Heat until it thickens into a paste.

3. Add Algae Pigment:

Mix algae powder with some water into a smooth paste.

4. Combine & Mix:

Blend the pigment with the binder to form algae paint.

5. Add Preservative:

A few drops of vinegar or clove oil can help the paint last longer.

For long-lasting algae paint, add 0.15% potassium sorbate (about 0.15 g per 100 ml of paint) and 10–15 drops of tea tree oil per 100 ml. Mix thoroughly to ensure even distribution. This combination helps prevent mold, bacteria, and spoilage for several months.





PROBLEM STATEMENT:

Conventional synthetic paints often contain harmful volatile organic compounds (VOCs) and non-biodegradable components that pollute indoor air and the environment. There is a need for an affordable, safe, and eco-friendly paint alternative for indoor wall decoration.

CONCLUSION:

OxyWall not only improves air quality but also promotes sustainability and teaches us about nature-based solutions.

It’s a simple step to make our classrooms greener, cleaner, and better for learning.